

FINAL PROJECT REPORT: HEART DISEASE PROBABILITY ESTIMATOR MODEL

Abstract

According to the World Health Organization (2025), heart disease remains a leading cause of global mortality, necessitating the development of more accurate and accessible diagnostic tools for early intervention. While traditional clinical check-ups provide essential snapshots of health, they often fail to capture the continuous physiological changes that signal declining cardiovascular function. Consequently, individuals with similar clinical profiles can exhibit vastly different underlying risks, highlighting the limitations of point-in-time assessments alone.

We suspect that, by integrating real-time physiological data with baseline indicators, we can identify high-risk events with significantly greater accuracy than static methods. Our approach involves: establishing a baseline via Logistic Regression on CDC health indicators, and integrating the MMASH (Multilevel Monitoring of Activity and Sleep in Healthy People) dataset through hourly time-series windowing. Then, using an XGBoost classifier, we would analyze the dynamic interplay between Heart Rate Variability (HRV), sleep fragmentation index, and user's physical activity levels.

While the static model achieved a ROC-AUC (Receiver Operating Characteristics - Area Under Curve) of 0.79 with a recall of only 0.02, the dynamic time-series model addition attained a ROC-AUC of 0.98 and a recall of 0.91, demonstrating a x45 improvement in detecting acute physiological strain.

Methods

This study uses a two-stage modeling approach to connect static clinical risk and dynamic physiological strain. The objective is to determine if real-time wearable metrics can provide an additional metric for heart disease than static survey data alone.

Data & Modeling Strategy

Stage	Dataset	Key Features & Pre-processing	Model & Rationale
Static Modeling	CDC Heart Disease Indicators (n = 253,680)	Features Used: Age, Sex, HighBP, HighChol, Diabetes, Smoker, Alcohol Consumption Cleaning: Deduplication, outlier filtering.	Logistic Regression: Chosen for high interpretability and its ability to output baseline probability scores
Dynamic Modeling	MMASH (Wearable Time-Series) (n = 22)	Features Used: Hour of Day, Physical Movement, Sleep Fragmentation, Weight, Age Processing: Hourly windowing (expanding 22 users into 543 snapshots) and Z-score scaling	Gradient Boosted Tree: Selected for its performance with high variance data and its ability to map non-linear physiological patterns

Data Cleaning & Pre-processing

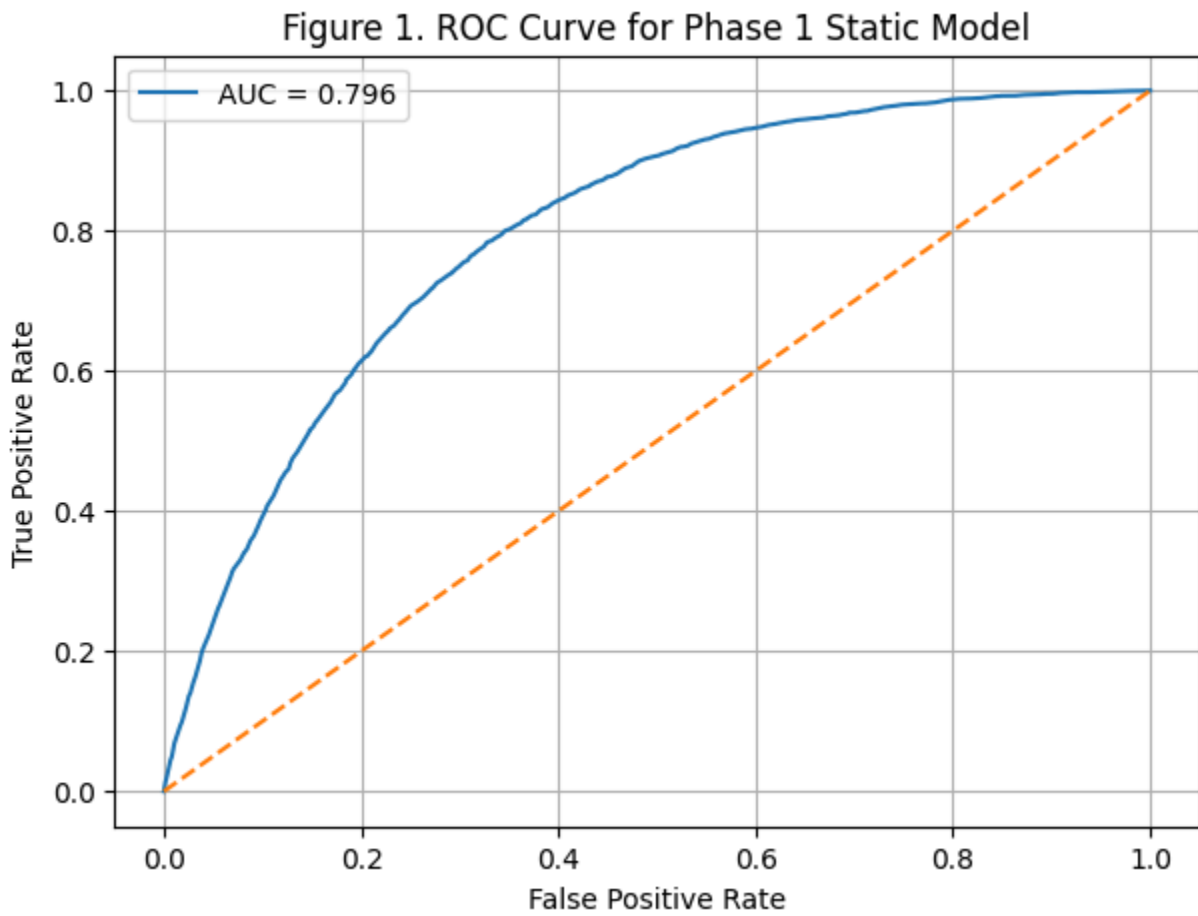
In phase 1, we deduplicated the Heart Disease Health Indicators Dataset by removing >23,000 rows and BMI outlier filtering was applied to reduce noise.

In phase 2, the raw MMASH dataset went through extensive hierarchical extraction. We chunked millions of raw beat-to-beat (RR) and actigraphy rows into hourly windows and calculated RMSSD (Heart Rate Variability) and Cardiovascular Strain to serve as proxy targets and merged these with sleep fragmentation index data using median imputation for missing values.

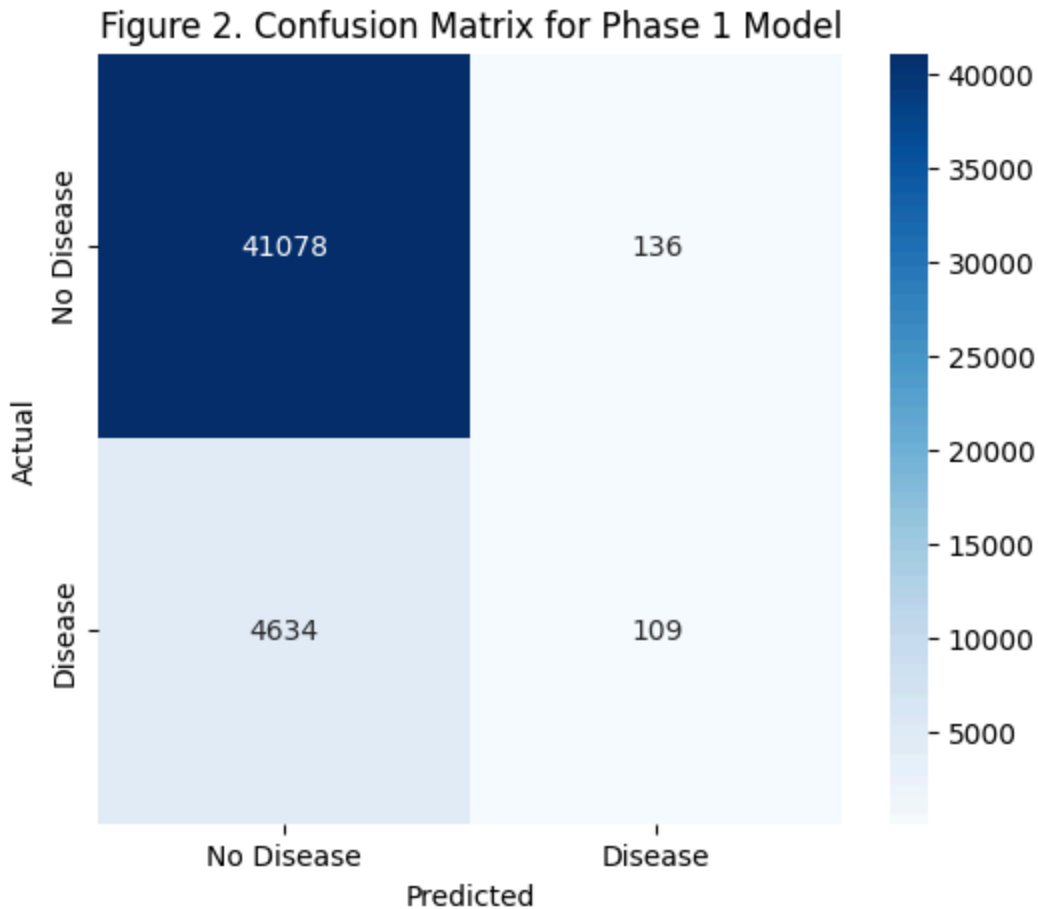
Assumptions

We assumed that high physiological strain (low HRV + high HR relative to movement) in healthy subjects serves as a valid proxy for the autonomic red flags that precede possible future cardiac events. We also assumed that 60-minute windowing provides a sufficient balance between signal density and computational efficiency for detecting significant shifts in cardiovascular state.

Results

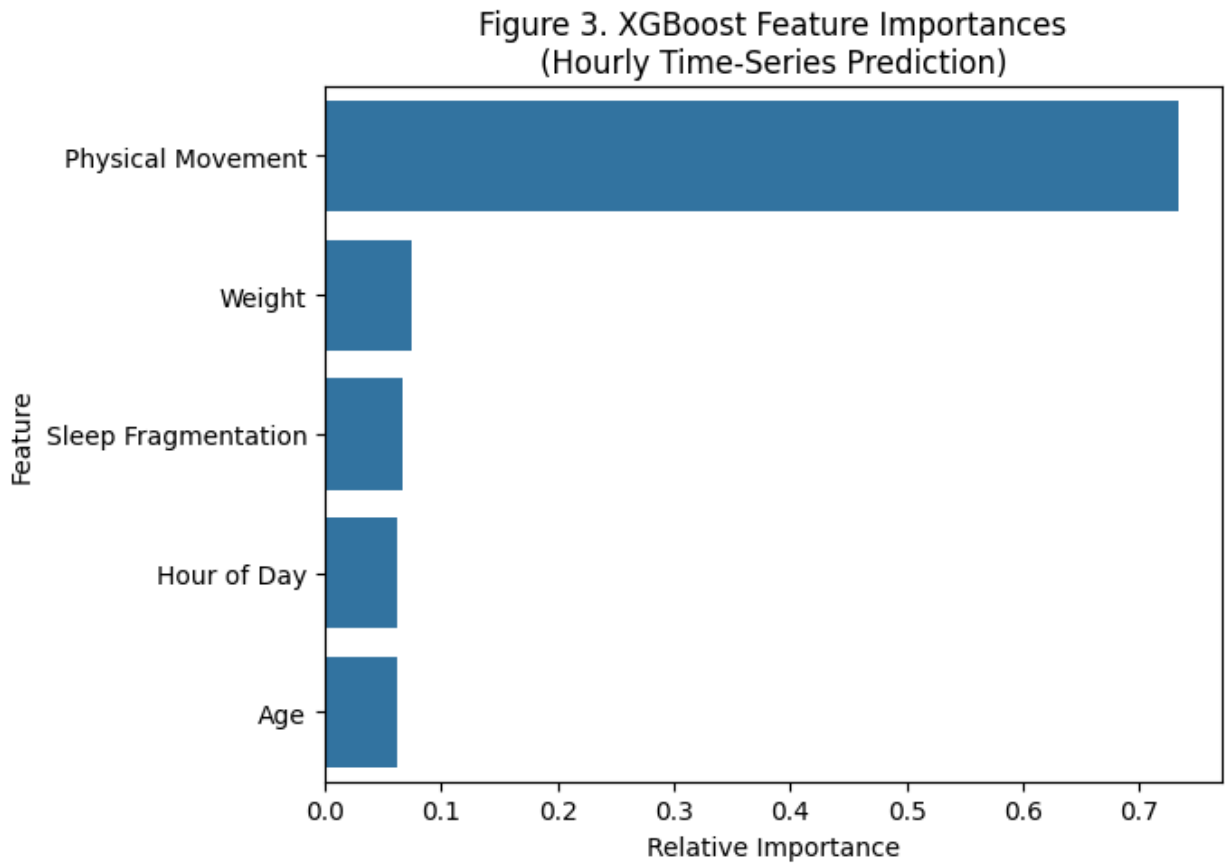


The initial logistic regression model utilized static biological markers and lifestyle factors. The model achieved an ROC-AUC score of 0.79, demonstrating a strong baseline ability to distinguish cardiovascular risk.

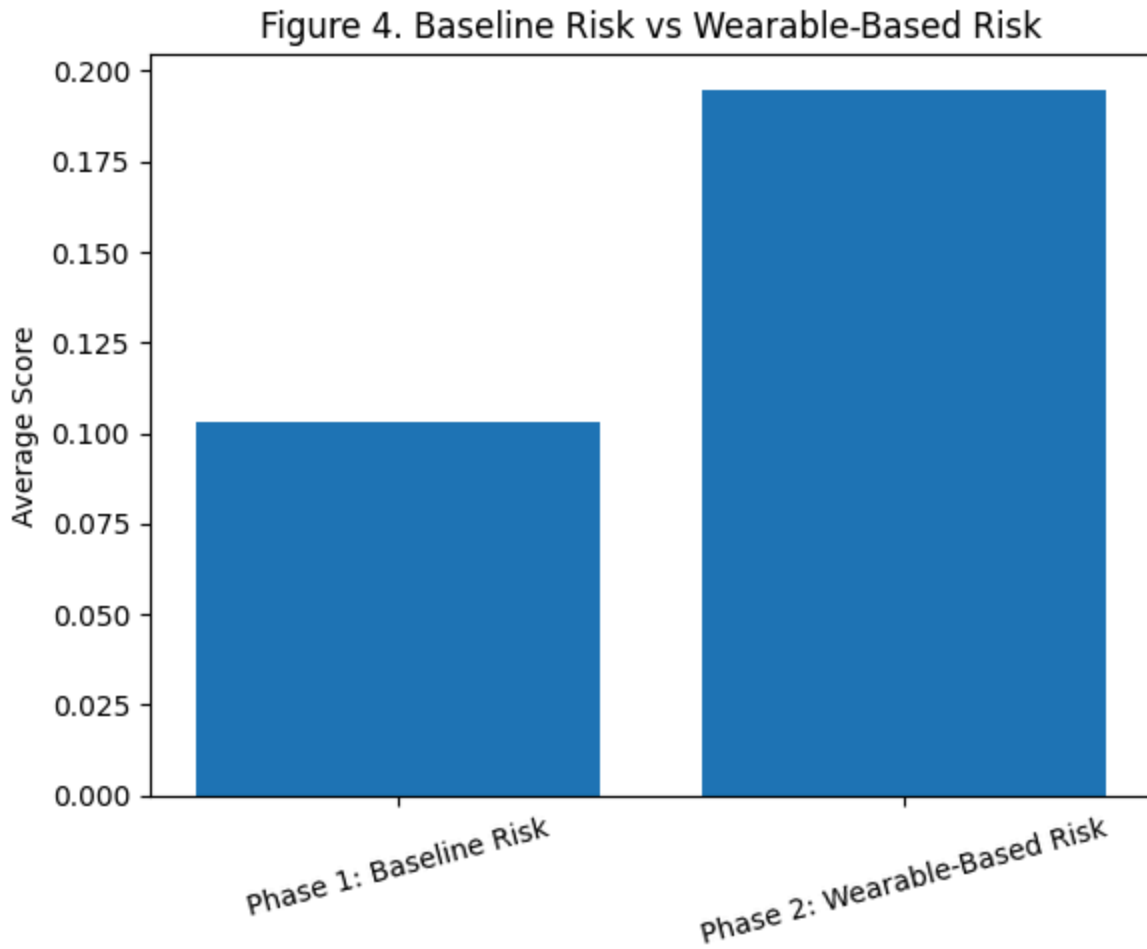


As shown in the confusion matrix (Figure 2), the model exhibited high precision (0.90) for the negative class but struggled with a low recall (0.02) for the positive class, indicating a tendency to miss at-risk individuals when relying solely on static data.

Integration of the MMASH dataset allowed for the training of a gradient boosted tree (XGBoost) on hourly physiological windows. This model significantly outperformed the logistic regression model, achieving an ROC-AUC score of 0.9838 and an overall accuracy of 95%.



As illustrated in the Feature Importance Plot (Figure 3), "Hour of Day" and "Physical Movement" were the primary drivers of the model's predictive power.



Finally, a comparison of risk probability scores (Figure 4) showed that the wearable-based model identified a higher average risk probability score (0.1948) compared to the baseline (0.1032), successfully capturing acute physiological strain that static surveys omitted.

Discussion

Interpretation of Results

The findings of this project highlight a disparity between static clinical profiling and dynamic physiological monitoring. While the phase 1 logistic regression achieved a ROC-AUC of 0.79, its 0.02 recall revealed a blind spot where static biomarkers like BMI and age are excellent for broad population studies but can be ineffective at flagging more acute cardiovascular events. By shifting to the phase 2 boosted tree model, the recall jumped to 0.91, representing a strong performance in detecting physiological red flags. For future work, expanding the dataset to include more diverse populations and integrating external factors such as diet, stress levels, and environmental conditions could improve the model.

We interpret these results as evidence that heart disease risk should not merely be defined by static demographics but rather a dynamic response to lifestyle stressors. The model's reliance on physical movement and sleep fragmentation over age and weight suggests that behavioral recovery metrics are also important indicators of short-term physiological strain. Practically, this system could be used in wearable applications to provide alerts, notifying users when their HRV and activity levels suggest a dangerous autonomic imbalance long before physical symptoms appear.

Reflections on the Process

The biggest challenge we faced was the complexity of the MMASH dataset, loading millions of rows across folders required not only standard CSV but also a recursive automation.

Furthermore, we encountered a major data leakage challenge during initial modeling. The first iteration using a random forest achieved a perfect 1.0 accuracy because it was overlooking other factors by reading the heart rate data that was used to define the target label. To overcome this, we re-engineered the feature set to be leak-free, forcing another model, booster tree, to predict strain using only context clues like movement and sleep.

Finally, the small sample size ($n = 22$) posed a threat to model generalizability. By implementing Time-Series Expansion, we successfully transformed the dataset into 543 unique hourly windows, providing the volume necessary for the XGBoost algorithm to identify significant patterns. This process was essential in moving from a correlation to a strong predictive tool.

Team Responsibilities

- Amelia: Datasets Discovery, Model Building, PowerPoints/Documentation
- Kiet: Data Pre-processing, Model Analysis, PowerPoints/Documentation
- Jenny: Model Analysis, Result's Visualization, PowerPoints/Documentation

Citations & Contributions

Teboul, A. (2021). [*Heart Disease Health Indicators Dataset*](#) [Data set]. Kaggle.

Rossi, A., et al. (2020). [*Multilevel Monitoring of Activity and Sleep in Healthy People*](#) (version 1.0.0). PhysioNet.

World Health Organization. (2025, July 31). *Cardiovascular diseases (CVDs)*. [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds))

AI Contributions Statement

Generative AI (Google Gemini) was used for the development of this project. The AI's contributions included:

- Developing the hierarchical data extraction and providing suggestions on how to transform raw sensor data into hourly physiological windows for the MMASH dataset.
- Helping in the transitions of the MMASH data into the static logistic regression using gradient boosted tree (XGBoost).